Chapter 5, Problem 7PP

Problem

Design a difference amplifier with gain 7.5.

Step-by-step solution

Step 1 of 4

Draw the difference amplifier circuit diagram as shown in Figure 1.

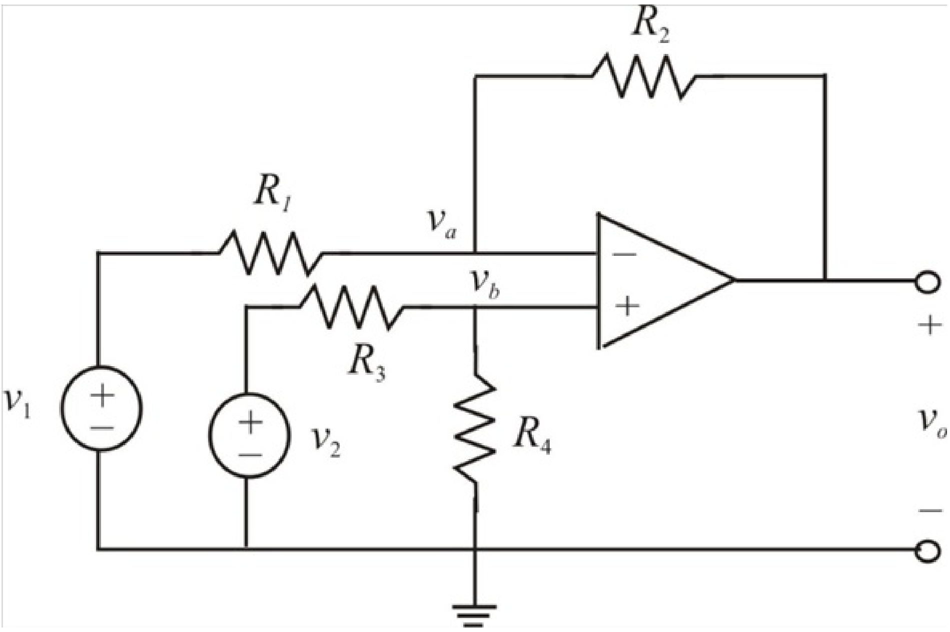

Figure 1

Step 2 of 4

The gain of the difference amplifier $= 7.5$

The circuit is labeled as shown in Figure 2.

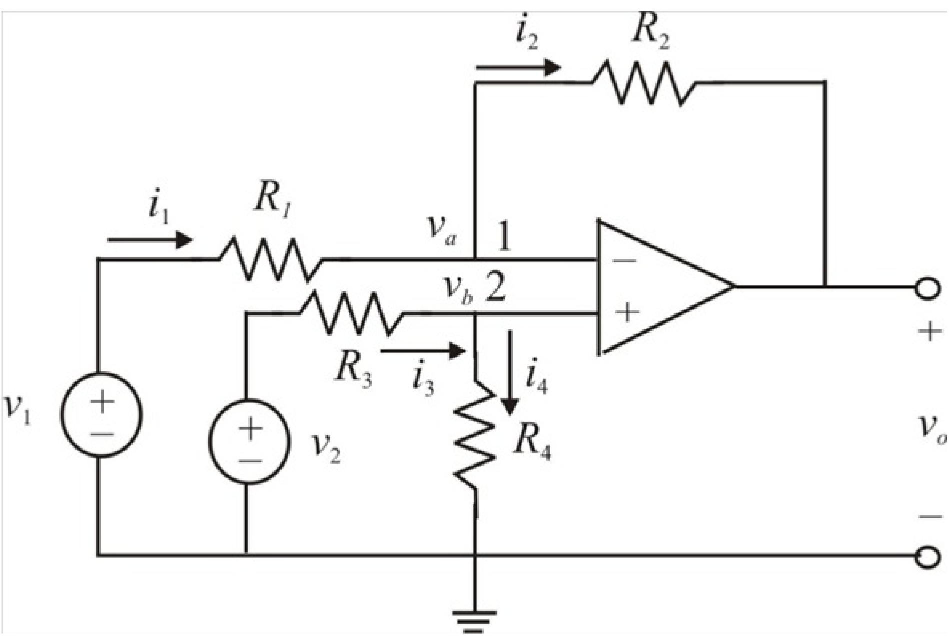

Figure 2

Step 3 of 4

Applying Kirchhoff's current law to node 1,

$$i_1 = i_2$$

$$\frac{v_1 - v_a}{R_1} = \frac{v_a - v_o}{R_2}$$

$$v_o = \left(\frac{R_2}{R_1} + 1\right) v_a - \frac{R_2}{R_1} v_1$$

Therefore $v_o = \left(\frac{R_2}{R_1} + 1\right) v_a - \frac{R_2}{R_1} v_1$ …… (1)

Applying Kirchhoff's current law to node 2,

$$i_3 = i_4$$

$$\frac{v_2 - v_b}{R_3} = \frac{v_b - 0}{R_4}$$

$$v_b = \frac{R_4}{R_3 + R_4} v_2$$

Therefore $v_b = \frac{R_4}{R_3 + R_4} v_2$ …… (2)

Step 4 of 4

But $v_a = v_b$ for an ideal op amp, substituting the equation (2) in equation (1), we get

$$v_o = \left(\frac{R_2}{R_1}+1\right)\frac{R_4}{R_3+R_4}v_2 - \frac{R_2}{R_1}v_1$$
$$= \frac{R_2\left(1+R_1/R_2\right)}{R_1\left(1+R_3/R_4\right)}v_2 - \frac{R_2}{R_1}v_1$$

Since a difference amplifier must reject a signal common to the two inputs, the amplifier must have the property that $v_o = 0$ when $v_1 = v_2$.

This property exist when

$$\frac{R_1}{R_2} = \frac{R_3}{R_4}$$

When the op amp circuit is a difference amplifier v_o becomes,

$$v_o = \frac{R_2}{R_1}\left(v_2 - v_1\right)$$

Hence the gain is

$$\frac{v_o}{v_2 - v_1} = \frac{R_2}{R_1} = 7.5$$
$$R_2 = 7.5\,R_1$$

But

$$\frac{R_1}{R_2} = \frac{R_3}{R_4} = \frac{1}{7.5}$$
$$R_4 = 7.5\,R_3$$

If $R_1 = 20\,\text{k}\Omega$, then $R_2 = 150\,\text{k}\Omega$

If $R_3 = 20\,\text{k}\Omega$, then $R_4 = 150\,\text{k}\Omega$

Therefore the resistance R_1 is $\boxed{20\,\text{k}\Omega}$.

Therefore the resistance R_2 is $\boxed{150\,\text{k}\Omega}$.

Therefore the resistance R_3 is $\boxed{20\,\text{k}\Omega}$.

Therefore the resistance R_4 is $\boxed{150\,\text{k}\Omega}$.

The desired difference amplifier circuit diagram as shown in Figure 3.

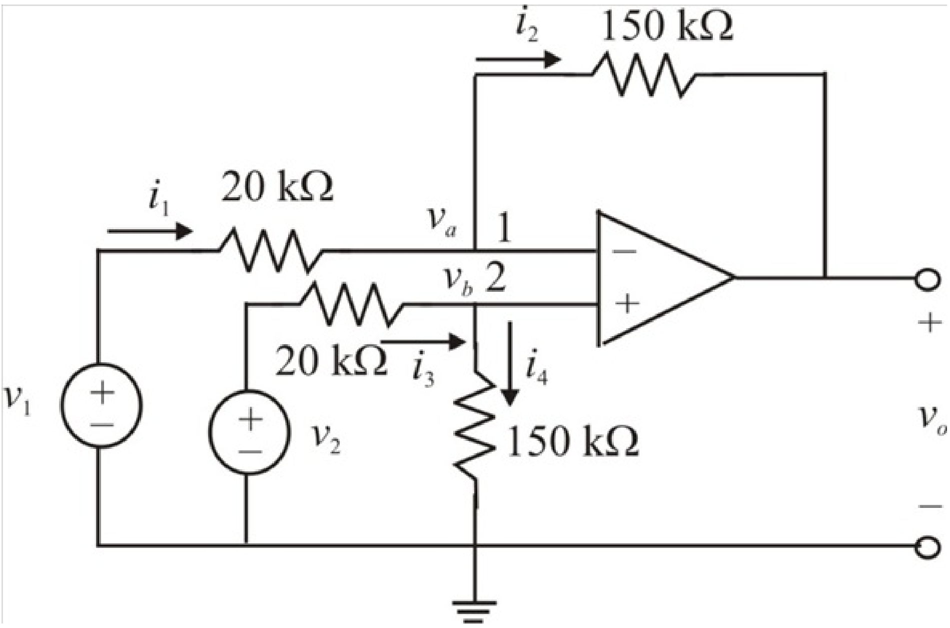

Figure 3